

2023 级 线性代数 IIB 参考答案

1. -11

2. $2x^2 + 7xy + 5y^2$

3. $\frac{1}{25} \begin{pmatrix} 3 & 4 \\ -4 & 3 \end{pmatrix}$

4. 27A

5. $\frac{9}{2}$

6. $\begin{pmatrix} 1 \\ -2 \\ 1 \end{pmatrix}$

7. A+3E

8. B 9. A 10. B 11. A 12. D

13 解:

$$\begin{vmatrix} \lambda & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = \begin{vmatrix} \lambda+2 & \lambda+2 & \lambda+2 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix} = (\lambda+2) \begin{vmatrix} 1 & 1 & 1 \\ 1 & \lambda & 1 \\ 1 & 1 & \lambda \end{vmatrix}$$

$$= (\lambda+2) \begin{vmatrix} 1 & 1 & 1 \\ 0 & \lambda-1 & 0 \\ 0 & 0 & \lambda-1 \end{vmatrix} = (\lambda+2)(\lambda-1)^2 \quad \dots 4'$$

① 当 $\lambda \neq -2$ 且 $\lambda \neq 1$ 时, 方程组有唯一解

② 当 $\lambda = 1$ 时:

$$(A|b) = \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$$\therefore r(A) = r(A|b) = 1 < 3$$

$\therefore$ 当 $\lambda = 1$ 时, 方程组有无穷多解, 通解为:

$$x = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + k_1 \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} + k_2 \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} \quad 4'$$

③ 当 $\lambda = -2$ 时,

$$(A|b) = \left(\begin{array}{ccc|ccc} -2 & 1 & 1 & 1 & 1 & 1 \\ 1 & -2 & 1 & 1 & -2 & -2 \\ 1 & 1 & -2 & 1 & 4 & 4 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & -2 & 1 & 4 & 4 \\ 0 & 3 & -3 & 0 & 9 & 9 \\ 0 & -3 & 3 & 0 & -6 & -6 \end{array} \right) \rightarrow \left(\begin{array}{ccc|ccc} 1 & 1 & -2 & 1 & 4 & 4 \\ 0 & 1 & -1 & 3 & 3 & 3 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{array} \right)$$

$$\therefore r(A) = 2 < r(A|b) = 3$$

$\therefore$ 当 $\lambda = -2$ 时, 方程组无解.

4. 解

$$\left(\begin{array}{ccccc} 1 & -2 & -4 & -1 & 5 \\ 0 & 1 & 3 & 1 & -2 \\ 1 & 1 & 5 & 3 & -2 \\ 2 & -1 & 1 & 4 & 1 \end{array} \right) \rightarrow \left(\begin{array}{ccccc} 1 & -2 & -4 & -1 & 5 \\ 0 & 1 & 3 & 1 & -2 \\ 0 & 3 & 9 & 4 & -7 \\ 0 & 3 & 9 & 6 & -9 \end{array} \right) \rightarrow \left(\begin{array}{ccccc} 1 & -2 & -4 & -1 & 5 \\ 0 & 1 & 3 & 1 & -2 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 3 & -3 \end{array} \right)$$

$$\rightarrow \left(\begin{array}{ccccc} 1 & -2 & -4 & -1 & 5 \\ 0 & 1 & 3 & 1 & -2 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccccc} 1 & -2 & -4 & 0 & 4 \\ 0 & 1 & 3 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \rightarrow \left(\begin{array}{ccccc} 1 & 0 & 2 & 0 & 2 \\ 0 & 1 & 3 & 0 & -1 \\ 0 & 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

$\therefore$ 向量组的秩为 3, 极大无关组取为 $\alpha_1, \alpha_2, \alpha_4$

$$\alpha_3 = 2\alpha_1 + 3\alpha_2,$$

$$\alpha_5 = 2\alpha_1 - \alpha_2 - \alpha_4$$

$$15. \begin{vmatrix} 2024-\lambda & 2024 & 2024 \\ 0 & -\lambda & 0 \\ 0 & 0 & -\lambda \end{vmatrix} = \lambda^2(2024-\lambda)$$

$$\therefore \lambda_1 = \lambda_2 = 0, \lambda_3 = 2024$$

4'

$$A - \lambda E = \begin{pmatrix} 2024 & 2024 & 2024 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \rightarrow \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\therefore p_1 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} \quad p_2 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$$

4'

$$A - \lambda_3 E = \begin{pmatrix} 0 & 2024 & 2024 \\ 0 & -2024 & 0 \\ 0 & 0 & -2024 \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

$$\therefore p_3 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

2'

$$\text{令 } P = (p_1 \ p_2 \ p_3), \text{ 则}$$

$$P^{-1}AP = \begin{pmatrix} 0 & & \\ & 0 & \\ & & 2024 \end{pmatrix} = \Lambda$$

2'

$$16. \text{证: 设 } k_1\alpha + k_2\beta + k_3\gamma = 0$$

$\because \alpha, \beta, \gamma$ 相互正交

$$\therefore \alpha, \beta, \gamma \neq 0, \text{ 且 } (\alpha, \beta) = (\beta, \gamma) = (\gamma, \alpha) = 0$$

$$\therefore (\alpha, k_1\alpha + k_2\beta + k_3\gamma) = k_1|\alpha|^2 = 0$$

$$\therefore k_1 = 0$$

$$\text{同理 } k_2 = 0, k_3 = 0$$

$$17. Ax = 0 \Rightarrow A^T A x = 0$$

$$\begin{aligned} A^T A x = 0 &\Rightarrow x^T A^T A x = 0 \Rightarrow (Ax)^T (Ax) = 0 \\ &\Rightarrow \|Ax\|^2 = 0 \Rightarrow Ax = 0 \end{aligned}$$

$\therefore$ 方程组 $Ax = 0$ 和 $A^T A x = 0$ 同解

$\therefore$ 它们的解空间的维数相同

$$\therefore r(A^T A) = r(A).$$